

PART I

General instructions. Fill in the information asked above. Write legibly your solutions and answers in the answer sheets provided. Submit this questionnaire along with the answer sheet.

Part I. For fill-in-the-blank questions, each blank is worth a point. For true-or-false questions, write only T or F. For identification questions, each desired item is worth a point.

1. The **free body diagram** of a rigid body represents it as having being isolated from its surroundings, replacing physical supports with forces and couple moments.
2. **True** or False. An ideal truss does not have three-force members.
3. For 3D systems, the internal moment has two components: **torsional** and **bending**, whose axes are tangent and normal to the cross section, respectively.
4. A body is said to be **under** constrained if it experiences a fewer number of reactive forces than there are applicable equations of static equilibrium.
5. True or **False**. A beam is subject to a load uniformly distributed throughout its length. The beam is supported at its left end by a pin, at its right end by a rocker, and at its midpoint by a roller. The beam is statically determinate.
6. For a simply supported thirteen-member plane truss to be considered simple, it must have **8** joints.

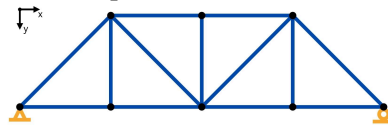


Figure 1.1

7. True or **False**. A frictionless ball-and-socket joint resists translations and rotations.
8. A body is considered **completely** constrained if all the reactive forces intersect at a common point or pass through a common axis, or if all reactive forces are parallel.
9. **True** or False. A stable body requires that the lines of action of the reactive forces do not intersect a common axis and are not parallel to one another.
10. For internal loadings of 2D systems, the **normal** force and the **shear** force are normal and tangent to the cross section, respectively.

Scenario 1. A 1.1-m long cantilever beam is subject to a 2-kN weight at its free end.

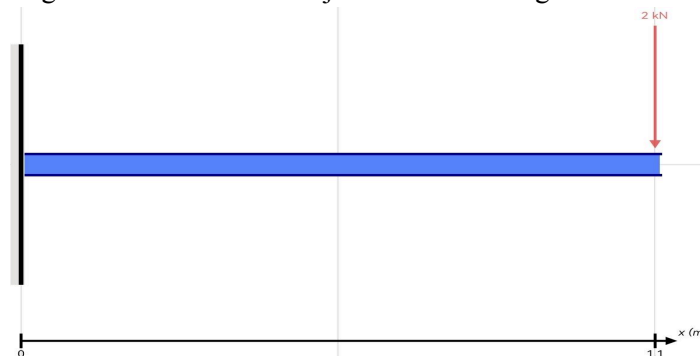


Figure 2.1

11. **True** or False. This scenario is statically determinate.
12. The support reactions have magnitudes of **2.0** kN and **2.2** kN-m.

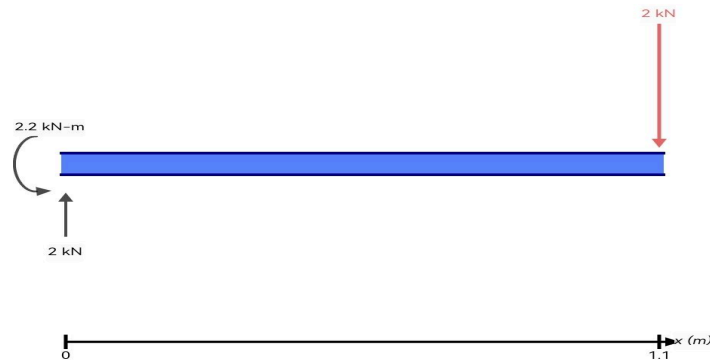


Figure 2.2

13. The maximum bending moment developed is 2.2 kN-m in magnitude and occurs 1.1 m from the free end.

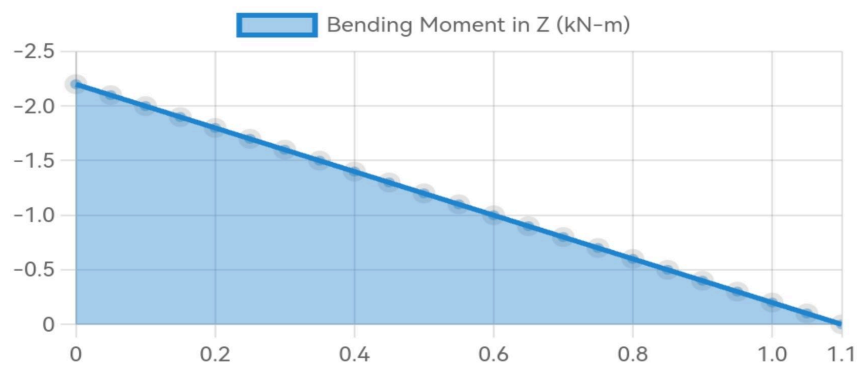


Figure 2.3

14. The maximum shear force developed is 2.0 kN-m in magnitude and occurs 0 m from the free end.

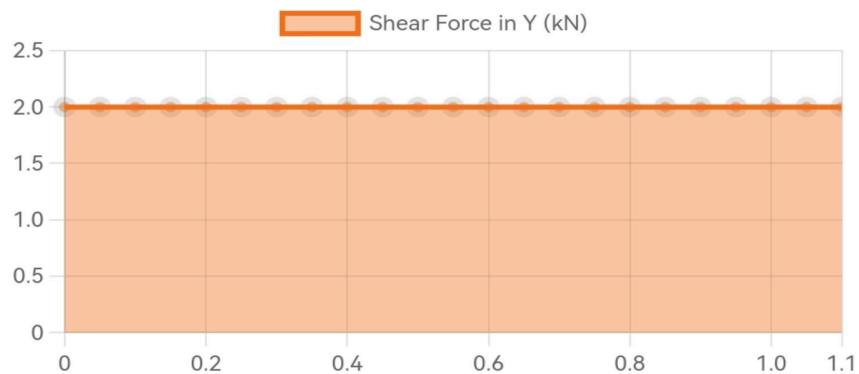


Figure 2.4

15. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree 0.
16. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree 1.

Scenario 2. A 1-m long beam carries a load uniformly distributed throughout its length. Assume a roller support 0.1 m from the left end and a pin support at the right end. The load has an intensity of 1 kN/m.

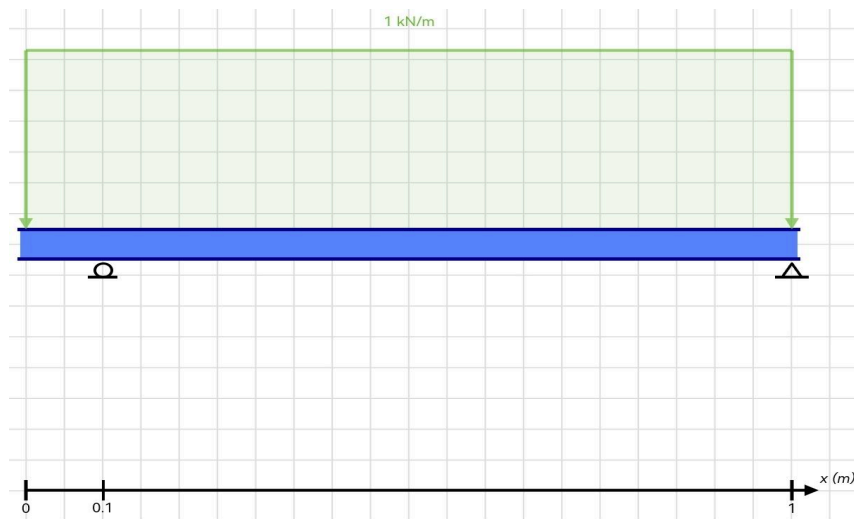


Figure 3.1

17. The roller support develops a force of **0.556** kN.

18. The pin support develops a force of **0.444** kN.

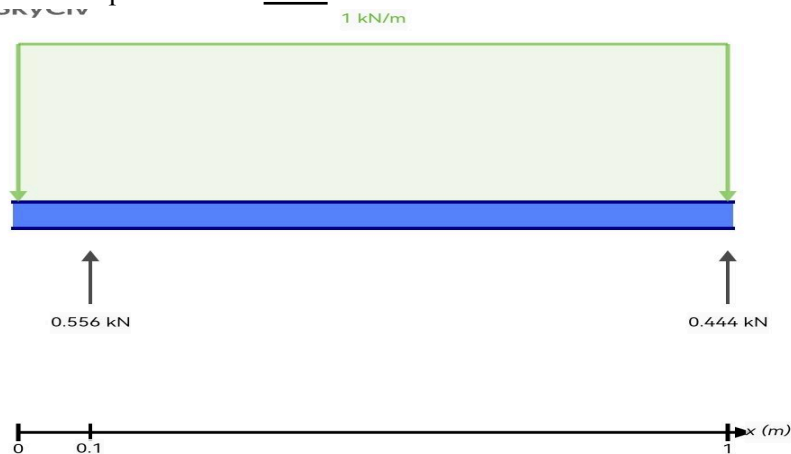


Figure 3.2

19. The bending moment developed at the right end is **0** kN-m.

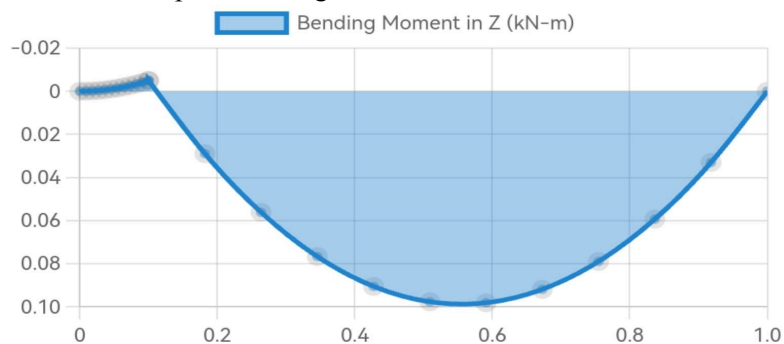


Figure 3.3

20. The shear force developed just to the left of the pin is **-0.444** kN.

21. The shear force developed just to the right of the pin is **0** kN.

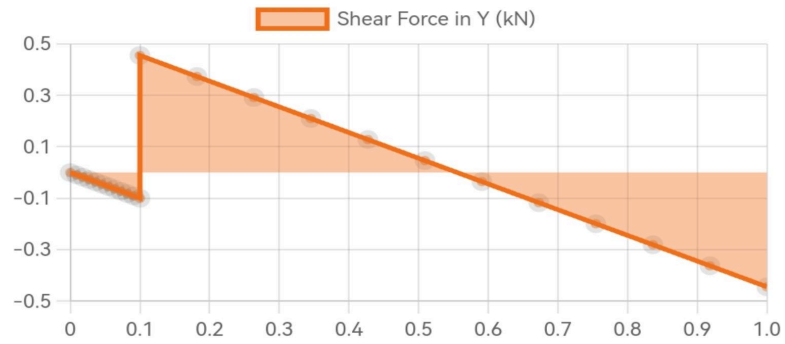


Figure 3.4

22. True or **False**. There is zero shear force developed at a point 0.9 from the right end.
23. The graph of the shear force as a function of location along the length of the beam is that of a polynomial of degree 1.
24. The graph of the bending moment as a function of location along the length of the beam is that of a polynomial of degree 2.

PART II

Where applicable, provide force diagrams and indicate the sense of member forces. Enclose final answers in boxes, with non-integer values rounded to three (3) decimal places.

PROBLEM 1

Consider the truss subject to idealized loads as shown above. Assume that the members have negligible weights and that the joints behave as frictionless pins. The magnitudes of p_1 , p_2 , and p_3 are 5 kN, 6 kN, and 7 kN, respectively. Set the dimensional parameters d and h to 2 ft and 3 ft, respectively.

A) **Find:** Forces in members FG, CF, and CD using the method of sections.

Given:

- ❖ $P_1 = 5$ kN, $P_2 = 6$ kN, $P_3 = 7$ kN, where P_1 , P_2 , P_3 are **load forces** applied at the truss joints (A,G,F).
- ❖ $d = 2$ ft, $h = 3$ ft, where d is the **width** of each panel in the truss and h is the vertical height of the truss.
- ❖ For the first section of the truss (from A $\rightarrow$ G $\rightarrow$ F), the structure has a **horizontal run of $2d = 4$ ft**, and a **vertical rise of $d = 2$ ft**. The corresponding slope triangle has: Horizontal = 2, Vertical = 1. Hypotenuse = $\sqrt{2^2 + 1^2} = \sqrt{5}$. Therefore, $\cos \theta = \frac{2}{\sqrt{5}} \approx 0.8944$, $\sin \theta = \frac{1}{\sqrt{5}} \approx 0.4472$

Solution:

Finding X and Y component of the load:

The loads P_1 , P_2 , P_3 are applied $\perp$ to the top chord. Each load is then broken into its horizontal (x) and vertical (y) components using the slope of the truss.

$$\vec{P}_1 = 5\left(\frac{1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{j}\right) = 2.236\hat{i} - 4.472\hat{j} \text{ kN}$$

$$\vec{P}_2 = 6\left(\frac{1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{j}\right) = 2.683\hat{i} - 5.367\hat{j} \text{ kN}$$

$$\vec{P}_3 = 7\left(\frac{1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{j}\right) = 3.130\hat{i} - 6.261\hat{j} \text{ kN}$$

A vertical cut is made through the members **FG, CF, and CD**, and the **left section of the truss**—consisting of joints **A, B, C, and G**—is analyzed

The moment about joint C is taken to find F_{FG} :

Taking moments about C removes the unknown forces F_{CD} and F_{CF} from the equation.

- ❖ **Load P_1 at joint A:** The horizontal distance from joint C to joint A is $2d = 4$ ft while the vertical distance is 0.

$$M_{P_1} = P_{1y}(4) = 4.472(4) = 17.888 \text{ kN}\cdot\text{ft (Counterclockwise)}$$

- ❖ **Load P_2 at joint G:** Joint G is located at (2,1) relative to joint A. Relative to joint C (4, 0), joint G is at $\Delta x = -2$, $\Delta y = 1$.

$$M_{P_2} = (P_{2y})(\Delta x) - (P_{2x})(\Delta y) = (-5.367)(-2) - (2.683)(1) = 10.734 - 2.683 = 8.051 \text{ kN}\cdot\text{ft (Counterclockwise)}$$

- ❖ **Member FFG at joint G:** The force acts along the top chord. The shortest (perpendicular) distance from point C to line AG is $d \perp = \frac{4}{\sqrt{5}} \approx 1.789$ ft. $M_{FG} = -1.789F_{FG}$.

$$\sum M_C = 17.888 + 8.051 - 1.789F_{FG} = 0 \Rightarrow F_{FG} = \boxed{14.500 \text{ kN (T)}}$$

Moment about Joint C to determine F_{CD} :

Intersection of F_{FG} and F_{CF} is at F.

- ❖ **Moment of P_1 about joint F:** $M_{P_1} = 22.360$ kN.ft (Counterclockwise).
- ❖ **Moment of P_2 about joint F:** $M_{P_2} = 13.417$ kN.ft (Counterclockwise).
- ❖ **Moment of F_{CD} :** The force acts horizontally, and the lever arm is equal to the vertical height of F = 2 ft. $M_{CD} = -F_{CD}(2)$.

$$\sum M_F = 22.360 + 13.417 - 2F_{CD} = 0 \Rightarrow F_{CD} = \boxed{17.889 \text{ kN (C)}}$$

Summation of y-components to determine F_{CF} :

$$\sum F_y = -9.839 + 14.500(0.4472) + F_{CF} = 0 \Rightarrow F_{CF} = \boxed{3.354 \text{ kN (T)}}$$

B) **Find:** The forces developed in members DF and EF by using the method of joints.

Solution:

- ❖ **Isolation of Joint F**, where an applied force P_3 is located at (3.130, -6.261), and forces member F_{FG} and F_{CF} are known tension, while forces F_{EF} and F_{DF} are unknown.

$$\sum F_x = 3.130 - 12.97 + 0.894F_{EF} + 0.707F_{DF} = 0 \quad \text{Equation 1}$$

$$\sum F_y = -6.261 - 6.48 - 3.354 + 0.447F_{EF} - 0.707F_{DF} = 0 \quad \text{Equation 2}$$

By simultaneous equations:

$$F_{EF} = \boxed{19.330 \text{ kN (T)}}$$

$$F_{DF} = \boxed{10.520 \text{ kN (C)}}$$

C) If the truss is to be idealized as simply supported by a pin and a roller, at which joints must the pin and the roller be placed to maintain stability of the loaded truss? Justify your answer.

- ❖ To maintain the truss stable as a simply supported structure, a **pin should be at joint D** and a **roller at joint E**, with the roller resisting horizontal motion.

Justification:

Translation: Inclined loads create horizontal forces, so the pin at D restrains movement in both x and y directions.

Rotation: Cantilever loads produce overturning moments. The roller at E must provide a horizontal reaction to form a force couple with D, resisting rotation and ensuring stability.

This setup makes the truss statically determinate and stable.

D) A contingency is considered wherein joint A is effectively unloaded. Identify zero-force members, if there be any, for this case.

Contingency: Joint A is unloaded ($P_1 = 0$).

- ❖ **Joint A:** Connects members AB and AG, which are non-collinear. With no external load applied, equilibrium at this joint requires that $F_{AB} = 0$ and $F_{AG} = 0$.

- ❖ **Joint B:** Connects AB, BC, and BG. Since $F_{AB} = 0$, horizontal equilibrium requires $F_{BC} = 0$. Vertical equilibrium requires $F_{BG} = 0$.

Therefore, the zero-force members are **AB, AG, BG, and BC**

E) Determine the forces developed in the supports at E and at D. Hint: Use the computed forces in members CD, DF, and EF.

Reaction at joint E:

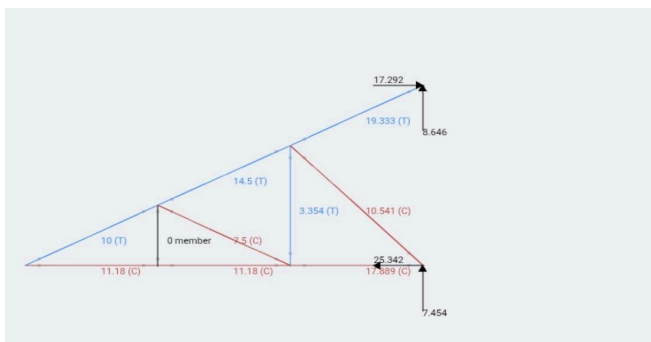
$$R_E = F_{EF} = \boxed{19.330 \text{ kN}}$$

Reaction at joint D:

$$R_{Dx} = F_{CD} + F_{DF} \cos(45^\circ) = 17.889 + 10.520(0.707) = 10.45 \text{ kN}$$

$$R_{Dy} = -F_{DF} \sin(45^\circ) = -10.520(0.707) = -7.44 \text{ kN}$$

$$|RD| = \sqrt{10.45^2 + 7.44^2} = \boxed{12.830 \text{ kN}}$$



The solution outlined above is concisely depicted in Figure 4, where blue indicates tension, red indicates compression, and black indicates supports.

PROBLEM 2

A custom beam design places supports at S_1 and at S_2 , in anticipation of a concentrated load of U_{point} kips and a uniformly distributed load of intensity u_{dist} kip/ft. Your main task is to analyze how the internal loadings induced vary as a function of x , where x is the distance from the left end of the beam. Consider the case when u_{point} is 5 kips and U_{dist} is 2 kip/ft. Set the respective values of $d_1, d_2, d_3, d_4, d_5, d_6$ to 3.2, 0.8, 2.0, 6.0, 2.0, and 0.6 ft, and those of m_v and m_h to 3 and 4. For static determinacy, assume a pin at S_1 and a roller at S_2 .

1. Problem Setup and Coordinate System

We establish a global coordinate system where x represents the distance from the left end of the beam. Using the provided parameters:

- Start of Beam: $x = 0$ ft
- Bracket Attachment Point (B): Located at
- Support S_1 (Pin): Located at $x_B + d_3 = 4.0 + 2.0 = 6.0$ ft.
- End of Distributed Load (D): Located at $x_{S1} + d_4 = 6.0 + 6.0 = 12.0$ ft.
- Support S_2 (Roller): Located at $x_D + d_5 = 12.0 + 2.0 = 14.0$ ft.

A) Determine the support reactions.

$$M_{\text{skip}} 5\left(\frac{3}{5}\right)(0.8) + 5\left(\frac{4}{5}\right)(0.6) = \boxed{4.8 \text{ kip.ft}}$$

$$\sum M_{S1}: 5\left(\frac{3}{5}\right)(2) + 4.8 + S_2(8) = 2(6) \left(\frac{6}{2}\right)$$

$$S_2 = \frac{36 - 4.8 - 6}{8} = \boxed{3.15 \text{ kip}}$$

$$\sum F_x \quad S_{1x} = 5\left(\frac{4}{5}\right) = \boxed{4 \text{ kip}}$$

$$\sum F_y \quad S_{1y} = 5\left(\frac{4}{5}\right) + 2(6) - 3.15 = \boxed{11.85 \text{ kip}}$$

B) Determine the normal force at $d_1 + d_2$ from the left end.

$$N = 5\left(\frac{4}{5}\right) = \boxed{4 \text{ kip}}$$

C) Determine the shear force at $d_1 + d_2$ from the left end.

$$V + 5\left(\frac{3}{5}\right) =$$

$$V = \boxed{-3 \text{ kip}}$$

D) Determine the bending moment at $d_1 + d_2$ from the left end.

$$4.8 + U = 0$$

$$U = \boxed{-4.8 \text{ kip.ft}}$$

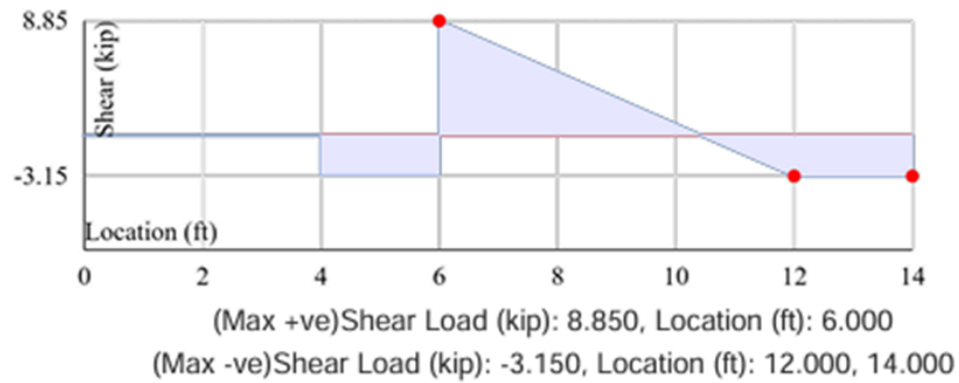
E) Determine the shear force as a function of x .

- $0 < x < 4$
 $n = 0$
 $v = 0$
- $4 < x < 6$
 $n = 4 \text{ kip}$
 $v = -3 \text{ kip}$
- $6 < x < 12$
 $n = 0$
 $v = -5\left(\frac{3}{5}\right) - 2(x - 6) + 11.85$
 $= 20.85 - 2x$
- $12 < x < 14$
 $v = -5(35) - 2(6) + 11.83$
 $= -3.15 \text{ kip}$

F) Determine the the bending moment as a function of x

- $0 < x < 4$
 $\Sigma M = 0$
- $4 < x < 6$
 $\Sigma M = -4.8 - 5\left(\frac{3}{5}\right)(x - 3)$
 $= 7.2 - 3x$
- $6 \leq x \leq 12$
 $\Sigma M = -4.8 - 5\left(\frac{3}{5}\right)(x - 4) - 2(x - 6)\left(\frac{x-6}{2}\right) + 11.85(x - 6)$
 $= -x^2 + 20.85x - 94.9$
- $12 < x < 14$
 $\Sigma M = -4.8 - 5(35)(x - 4) - 2(6)(x - 9) + 11.85(x - 6)$
 $= 44.1 - 3.15x$

- G) Plot the shear force diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations.



- H) Plot the bending moment diagram over the length of the beam, making sure to indicate the extrema, the zero points, and their locations.

